Research in International Business and Finance 79 (2025) 103099

Contents lists available at ScienceDirect

Research in International Business and Finance

journal homepage: www.elsevier.com/locate/ribaf

Institutional investor information network and corporate risk taking

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ARTICLE INFO

Keywords:
Institutional investor
Information sharing
Risk taking
Information asymmetry
Herding behavior of analysts

ABSTRACT

Institutional investors form the network through common heavy holdings, and there is information sharing in the network. This paper examines the impact of institutional investor information sharing on corporate risk taking. It is found that the information sharing of institutional investors can bring about the effect of external supervision and resource advantage, and significantly improve the level of enterprise risk taking. The mechanism analysis shows that the information sharing of institutional investors can improve the level of corporate risk taking mainly by reducing the information asymmetry and restraining the herding behavior of analysts. Further analysis shows that in the enterprises with high degree of financialization, high degree of financing constraint, no political connection, high agency cost, good internal control quality, good legal environment, and growth stage company, information sharing has a greater effect on the improvement of enterprise risk taking. This paper has enriched the study of social network and corporate risk taking, and has positive practical significance for enterprises to improve the efficiency of investment decision-making.

1. Introduction

In recent years, China's economy has been in a critical transition period from rapid growth to high-quality development. As the micro-foundation of the economy and society, enterprises not only contribute to the rapid development of the economy, but also contribute to the high-quality development of the economy. Enhancing the ability of enterprises to cope with future uncertainties and improving risk-taking levels play an important role in the high-quality development of the economy. The risk-taking level reflects the price a company willing to pay to obtain potentially high returns, which is reflected in the company's choice of investment projects that have a positive future net present value but are full of risks (Acharya et al., 2011; Boubakri et al., 2013). A higher level of risk-taking can enhance the long-term competitive advantage of enterprise and corporate value (Koirala et al., 2020; Cucculelli and Ermini, 2012), and also accelerate the accumulation of social capital as well as promoting long-term economic growth (John et al., 2008). Therefore, if an enterprise chooses investment projects with lower risks in the long term, both the enterprise's own production efficiency and the development of social productivity will be hindered. The level of risk-taking often depends on the management's risk propensity in operating activities and the company's ability to obtain resources. Based on the existing literature, the factors affecting corporate risk-taking mainly involve social environment and policy systems (Gopalan et al., 2021; Raithatha and Popli, 2022), corporate

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https://doi.org/10.1016/j.ribaf.2025.103099
Received 12 September 2024; Received in revised form 17 May 2025; Accepted 14 August 2025
Available online 15 August 2025
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micro-governance (Lin et al., 2019) and manager characteristics (Do et al., 2022). But what is more subjective and proactive is closely related to the management of risk decision-making entities. In the situations of principal-agent and information asymmetry in enterprises, management often avoids high-risk projects for the sake of their own compensation and professional reputation, resulting in some high-risk investment projects with positive future net present values being abandoned by management, which damages shareholder value and affecting macroeconomic growth. So, how to better supervise management behavior and improve corporate risk-taking is of great significance.

Professional institutional investors have the dual advantages of information acquisition and capital scale. They can conduct detailed collection and analysis of corporate information (Jiang and Yuan, 2018), and participate in corporate activities by soliciting voting proxies from small shareholders and submitting shareholder proposals (Matvos and Ostrovsky, 2010). So institutional investors are given high expectations in terms of corporate governance. However, market survey reports as of 2022 show that the participation of institutional investors and shareholders in corporate governance in China is still not ideal. Faced with many restrictive factors such as the excessive voice of corporate major shareholders and information asymmetry, institutional investors have higher cost in corporate governance. Therefore, individual institutional investor plays relatively limited role in corporate governance (Shleifer and Vishny, 1986). However, the joint participation of institutional investors in corporate governance may make up for the weak governance effectiveness of individual institutional investors and strengthen the governance effect on the company. Social embeddedness theory points out that any individual in economic society is in a certain relationship, and their behavioral decisions will be affected by the network. The same is true when it comes to asset selection. As peers, institutional investors frequently exchange information, and their decision-making behaviors will influence each other (Pool et al., 2015). The consistency of institutional investors' decision-making behavior will promote the performance of their functions at the group level (Cvijanović et al., 2022), which may make up for the shortcomings of individual institutional investors and improve the governance effect of enterprises (He et al., 2019).

Different institutional investors will allocate assets to the same listed company, resulting in the phenomenon of common shareholding. Based on this common shareholding behavior, an institutional investor network is formed among institutional investors, and there is information exchange and sharing in the network (Krishnamurti et al., 2021; Gong and Liu, 2023; Li et al., 2023). Therefore, this paper uses the joint shareholding behavior of institutional investors to construct an institutional investor network and studies the impact of information sharing in the network on corporate risk-taking. Currently, some scholars have studied the impact of different social networks on corporate risk-taking levels based on the perspective of social networks (Fan et al., 2021; Ferris et al., 2019), but there is no literature that directly studies the impact of institutional investors on corporate risk-taking levels from the network level.

The contributions of this paper mainly include the following aspects: (1) This paper enriches the literature on the factors affecting corporate risk-taking. Based on social network theory, this paper constructs an institutional investor network and finds that information sharing within the institutional investor network has external supervision and resource advantage effects, which helps to improve the level of corporate risk-taking, and expands and deepens the relevant research on corporate risk-taking from the social network level. (2) This paper enriches the research literature on the economic consequences of institutional investors. Based on the perspective of corporate risk-taking, we reveal the impact of the relationship network that is prevalent in my country's traditional relationship-based society on corporate governance. (3) This paper explores the differences in the role of institutional investor information sharing on corporate risk-taking behavior under different scenarios of financialization degree, financing constraints, political connections, agency costs, internal control quality, regional legal environment and life cycle.

2. Literature review and research hypothesis

2.1. Institutional investor network and corporate risk-taking

Risk selection is an important decision in the corporate investment decision-making process and is related to the company's risk-taking level. Tobin's theoretical analysis shows that in a perfect market, the real investment level can be predicted by marginal Q, and the company's investment behavior depends only on the investment opportunities represented by marginal Q. That is, managers should choose all investment projects with positive expected net present values so that marginal Q equals 1. However, in reality, managers' choice of investment projects is affected by many factors. According to the agency theory, since managers' human capital is a non-diversified investment and managers have no other investments to disperse risks, managers tend to abandon high-risk investment projects to avoid the personal wealth loss, unemployment risk and reputation loss. Hence, in the face of mismatch between income and required risk responsibilities, avoiding investment projects of higher risks but higher income becomes the logical choice. Especially when the contract between owners and managers is incomplete and information asymmetry occurs, management lacks corresponding constraints and often chooses to live a "quiet life" (Bertrand and Mullainathan, 2003), instead of actively choosing high-risk and high-yield investments project. In addition, enterprise risk-taking is a resource-consuming activity, and has strong resource dependence (Almeida and Campello, 2007). The investment projects and funds required for corporate investment require high information costs and transaction costs. This means that even if corporate managers are willing to engage in high-risk and high-return investment activities, they may lack the corresponding resources, resulting in insufficient investment and low investment efficiency (Faccio et al., 2016). Therefore, if a company cannot obtain sufficient resources, it is often difficult to make risky strategic investments, which in turn affects the company's risk-taking level.

Existing research shows that there is information sharing within institutional investor information networks (Krishnamurti et al., 2021; Gong and Liu, 2023; Li et al., 2023). Information sharing among institutional investors can produce two effects: one is the external supervision effect, the other is the resource advantage effect. First of all, information sharing among institutional investors can broaden the channels for institutional investors to obtain information, make up for the lack of information, and improve the

comprehensiveness of information obtained by institutional investors. Secondly, institutional investors can also verify the information and correct cognitive biases through direct information exchange to improve the accuracy of the information they obtain. The improvement in the comprehensiveness and accuracy of information obtained by institutional investors has greatly reduced the supervision costs of institutional investors' supervision of enterprises, prompting institutional investors to actively supervise corporate behaviors. In addition, the willingness of individual institutional investors to actively supervise management may be limited due to their weak voice, while institutional investor information sharing makes the behavior of institutional investors within the network consistent, prompting institutional investors to act together to exert greater influence on corporate governance. Therefore, information sharing among institutional investors can enable them to better play a supervisory role, and reduce opportunistic behavior of corporate management, thereby improving the level of risk taking.

Information sharing among investors on the network enables individual institutional investors to obtain more information on industry development, thereby increasing the richness and diversity of information to enterprises, which enable management to keep abreast of industry dynamics and development trends, and to discover more favorable investment opportunities more accurately and timely, thereby reducing the risk of underinvestment caused by management's lack of key decision-making information. At the same time, the resource allocation effect of the institutional investor network can more conveniently help companies obtain more resources needed for investment at a lower cost (Dahl and Pedersen, 2005). Based on the above analysis, this paper proposes hypothesis H1:

H1. Information sharing among institutional investors can improve corporate risk-taking levels.

2.2. Institutional investor network, information asymmetry and corporate risk-taking

The risk-taking behavior of the enterprise is highly dependent on resources, and the resources required include investment projects, funds, technology, etc. If there are not enough investment projects and cannot obtain enough funds, enterprises will face greater resource constraints when investing, resulting in lower corporate risk-taking levels. Investors' investment decisions will be affected by the information they obtain (Travis, 2021), Biddle et al. (2009) pointed out that information asymmetry is an important reason for corporate underinvestment and low investment efficiency. On the one hand, when the degree of information asymmetry increases, investors will reduce investment demand, which can reduce the investment projects available to the enterprise. On the other hand, higher information asymmetry will also increase corporate financing constraints. Bank loans are the main source for enterprises to obtain long-term development funds. Higher information asymmetry will increase banks' risk aversion motivation, banks will require higher loan interest rates and formulate more stringent loan treaties from enterprises based on the consideration of preventing loan risks, making it more difficult for companies to obtain external financing (Bai et al., 2006).

The external monitoring effect generated by information sharing by institutional investors can prompt managers to adopt better accounting policies, and institutional investors can also disclose the earnings management methods of their holding companies, thereby reducing the degree of information asymmetry between the outside world and enterprises (Chung et al., 2002). Supervision by institutional investors can also improve the quality of corporate information disclosure (Boone and White, 2015) and reduce the degree of information asymmetry between companies and external stakeholders. When institutional investors discover that corporate managers are hiding information or engaging in improper behavior, they can sue the managers, prompting them to proactively improve the quality of information disclosure (Cheng et al., 2010; Pukthuanthong et al., 2017). With reduced information asymmetry, external investors such as banks can obtain more corporate information, reduce investment risks, and make it easier for companies to obtain funds and investment projects. Based on the above analysis, this paper proposes hypothesis H2:

H2. Institutional investor information sharing improves corporate risk-taking by reducing information asymmetry.

2.3. Institutional investor network, herding behavior of analysts and corporate risk-taking

Securities analysts are important information intermediaries in the capital market, and investors can obtain relevant information about listed companies through research reports issued by securities analysts. However, there is obvious convergence in the behavior of analysts, and they will issue similar forecast ratings to other analysts, that is, there is analyst herding behavior (Welch, 2003). Analyst herding behavior will reduce corporate risk-taking levels. On the one hand, analysts' herding behavior reduces private information in research reports, resulting in less corporate information available in the market, which makes it more difficult for investors to understand the true situation of the company and evaluate the future returns of the company's high-risk projects. So, in order to avoid risks, investors will reduce the supply of funds, thereby increasing the financing constraints faced by enterprises and reducing their risk-taking levels. On the other hand, analyst herding behavior shows that many analysts simply imitate other people's research reports to obtain information about the company, and do not pay too much attention to the company's operating conditions, thereby reducing the external supervision pressure on the company's management, which induces opportunistic behavior of management, that is, the management invests in more low-risk projects for its own interests.

Existing researches have shown that there is a correlation between analysts' forecasting behavior and institutional investors, which proves that there is close exchange of information between institutional investors and analysts (Liu and Chen, 2024). Close communication between institutional investors and analysts will increase analysts' private information, which will reduce analyst herding behavior. The more frequent the exchange of information in the institutional investor network and the higher the degree of information sharing, the more private information analysts obtain, thereby reducing analyst herding behavior. Based on the above analysis, this paper proposes hypothesis H3:

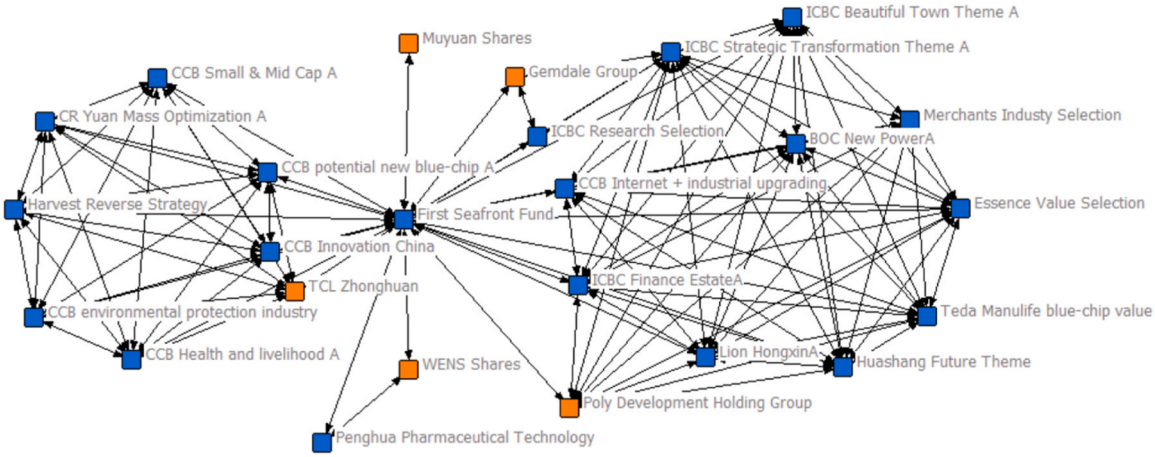

Fig. 1. The network of holding a large number of shares.

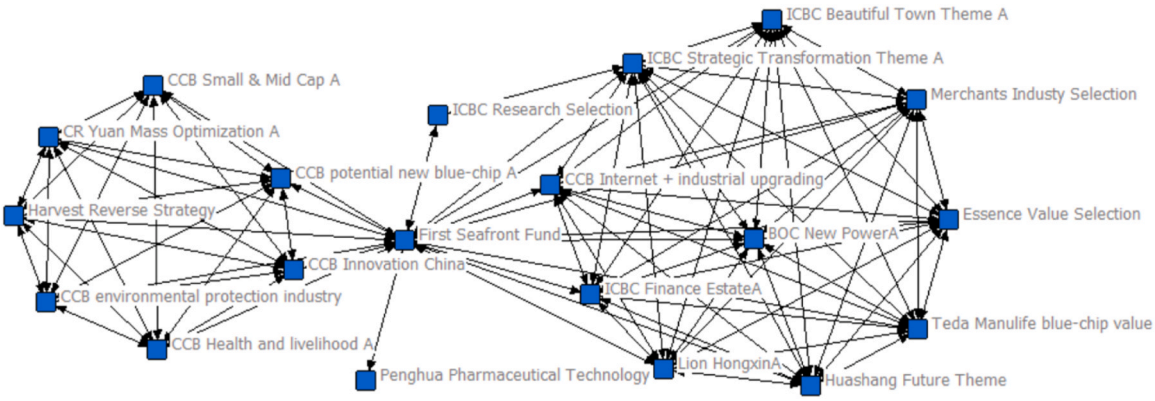

Fig. 2. The institutional investors network of Muyuan Shares.

H3. Institutional investor information sharing improves corporate risk-taking by inhibiting analysts herding behavior.

3. Research design

3.1. Institutional investor network

It is found that institutional investor networks are generally constructed through three methods. The first type is a network built based on the social relationships of institutional investors, such as alumni relationships (Galariotis et al., 2015). Fund managers in the same social network have similar shareholding behaviors and trading styles (Cici et al., 2017). And the second kind is a network based on geographical location, for example, fund managers living in the same city exhibit consistent trading behavior (Hong et al., 2005). The third kind is a network built based on asset allocation. Crane et al. (2019) studied the situation of the U.S. capital market, constructed an institutional investor network based on common ownership relationships, and then analyzed the corporate governance effect of the institutional investor network. The network built based on the shareholdings of institutional investors supports information exchange through the consistency of transactions and positions within the network, so the consistency of trading behavior among funds supports the existence of the network. Following the third method, this paper constructs an institutional investor network based on institutional investors' heavy holdings of stocks of the same listed company.

If institutional investors hold a large number of the same stocks, it is assumed that there is a connection between the two institutional investors. The institutional investor network is defined as a collection of funds that directly and jointly hold a heavy position in the stock (In the network, each institutional investor is a node, and two institutional investors are connected by a straight line.). The standard for large holdings is that the proportion of institutional investors' holdings accounts for 5 % or more of the fund's net value. As shown in Fig. 1, Muyuan shares are taken as an example. The blue squares in the picture represent funds, and the orange squares represent stocks. Muyuan shares were heavily held by Qianhai Kaiyuan Refinancing Theme Select Fund in mid-2021. At the same time, the fund also held heavy positions in four stocks including TCL Zhonghuan, Wen's Shares, Gemdale Group and Poly Development.

Table 1
Variable definition and description.

Variable name	Variable symbol	Variable meaning and description
Explanatory variables	*Density*	Network density. The larger the value, the higher the efficiency of information sharing.
	Center	Network centrality. The larger the value, the higher the efficiency of information sharing.
Explained variable	*Risk*	Risk-taking level. The larger the value, the higher the risk-taking level.
Mediating variable	*Asy*	KV measurement method, the smaller the value, the better the information disclosure quality and the smaller the degree of information asymmetry.
	Herd	Analyst herding behavior. The larger its value, the higher the degree of analyst herding behavior.
Control variables	*Size*	Enterprise size. The value is natural logarithm of total assets at the end of each year.
	Lev	Asset-liability ratio. The value is the ratio of total liabilities to total assets at the end of the period.
	Top1	Shareholding ratio of the largest shareholder.
	Salary	Salary sensitivity. Average salary of directors, supervisors and senior executives.
	TobinQ	Corporate performance.
	ROA	Return on assets.
	Age	Business operating life.
	Growth	Sales revenue growth rate. The value is the annual growth rate of corporate operating income.
	PPE	Fixed assets ratio. The value is the ratio of fixed assets to total assets at the end of the period.
	Cap	Capital expenditures. The value is the natural logarithm of the cash paid by an enterprise to build fixed assets, intangible assets and other long-term assets.
	Eps	Earnings per share.
	Dual	The two positions are combined into one. If the chairman and general manager are one person, the value is 1, otherwise it is 0.

Table 2
Descriptive statistics.

Variable	N	Mean	Sd	Min	Max	P25	P50	P75	P90
Risk	9515	0.0272	0.0281	0.0017	0.1699	0.0104	0.0182	0.0328	0.0579
Density	9415	0.4167	0.1904	0.1039	1	0.2802	0.3763	0.5119	0.6786
Center	9515	−0.2975	0.1565	−0.7180	0	−0.4198	−0.3042	−0.1655	−0.1000
Size	9515	22.7415	1.3861	20.3527	27.0485	21.7442	22.5285	23.5121	24.6088
Lev	9515	0.4201	0.1940	0.0569	0.8455	0.2655	0.4160	0.5656	0.6848
Top1	9514	34.2218	14.9071	8.4500	74.3000	22.5100	32.1600	44.0700	55.3000
Salary	9493	12.9354	0.7095	11.2938	14.9295	12.4553	12.9071	13.3642	13.8469
TobinQ	8996	2.5272	1.6548	0.8825	9.7768	1.4446	2.0142	3.0001	4.5565
ROA	9457	0.0310	0.0335	−0.1684	0.5733	0.0088	0.0230	0.0450	0.0723
Age	9514	2.8719	0.3196	1.9459	3.4965	2.7081	2.8904	3.0910	3.2581
Growth	9513	0.2539	0.4022	−0.3982	2.4463	0.0439	0.1707	0.3486	0.6266
PPE	9170	0.2017	0.1534	0.0032	0.6748	0.0839	0.1666	0.2827	0.4236
Cap	9513	19.4082	1.6866	15.1332	24.1042	18.3311	19.3500	20.4109	21.5949
Eps	9515	0.7550	0.9048	−1.1375	5.8816	0.2513	0.5293	0.9914	1.6629
Dual	9415	0.2908	0.4542	0	1	0	0	1	1

Among them, TCL Zhonghuan, Wen's Gemdale Holdings, Gemdale Group and Poly Development are heavily held by 19 funds including Harvest Reverser Strategy, Huashang Future Theme, etc. These 19 funds form a joint shareholding relationship with the CR Yuan Mass Optimization. Fig. 2 shows the institutional investor network of Muyuan Foodstuff Co., Ltd. consisting of Qianhai Kaiyuan Refinancing Theme Selection and other 19 funds.

3.2. Variable definition and measurement

Network density is an indicator that describes the structural characteristics of a network, and can effectively measure the rate of information flow within the network. The greater the network density, the faster the information flow (Pareek, 2011). Network centrality better describes the differences in the positions of different individuals in the network, and can effectively capture an individual's ability to obtain information and influence the behavior of other individuals (El-Khatib et al., 2015). Individuals with stronger network centrality have stronger abilities to obtain and transmit information through the network (Bajo et al., 2016). Therefore, this paper uses network density and network centrality to characterize the network characteristics of institutional investors, and measures the efficiency of information sharing from two dimensions: the speed of information dissemination in the network and the degree of control of information in the network by institutional investors.

3.2.1. Network density

Network density measures the closeness of the connections between institutional investors in the network. In the network, each institutional investor is a node, and two institutional investors are connected by a straight line. Network density is defined as the ratio of the number of edges actually connected between nodes in the network to the maximum number of edges that may exist. The specific

Table 3
Baseline model regression results.

	(1) Risk	(2) Risk	(3) Risk	(4) Risk
Density	0.0042***	0.0048***		
Center	(2.83)	(3.01)	0.0053***	0.0065***
			(3.11)	(3.54)
Size		−0.0006		−0.0005
		(−1.03)		(−0.93)
Lev		0.0031		0.0030
		(1.28)		(1.25)
Top1		−0.0001***		−0.0001***
		(−5.32)		(−5.26)
Salary		0.0006		0.0006
		(0.91)		(0.95)
TobinQ		0.0016***		0.0016***
		(6.50)		(6.58)
ROA		0.0061		0.0065
		(0.55)		(0.58)
Age		0.0021		0.0019
		(1.46)		(1.34)
Growth		0.0025***		0.0026***
		(3.60)		(3.63)
PPE		−0.0054*		−0.0051
		(−1.68)		(−1.61)
Cap		−0.0014***		−0.0014***
		(−3.93)		(−3.95)
Eps		−0.0017***		−0.0017***
		(−3.82)		(−3.76)
Dual		0.0008		0.0009
		(1.01)		(1.14)
Constant	0.0252***	0.0527***	0.0285***	0.0554***
	(4.09)	(4.27)	(4.65)	(4.57)
Ind	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	9415	8428	9515	8513
R^2	0.0563	0.0634	0.0549	0.0619

calculation method is shown in formula (1).

$$Density_i = 2E_i/k_i(k_i - 1) \tag{1}$$

Among them, $E_i$represents the actual number of edges in the network, k_i represents the number of nodes in the network, and$k_i(k_i - 1)/2$ is the maximum number of edges that may be connected between institutional investors.

The greater the network density, the closer the connections between institutional investors, the faster the information is transmitted within the network, and the higher the efficiency of information sharing.

3.2.2. Network centrality

The network centrality $Center_D$ is defined as the weighted sum of the differences between the network centrality $V_{\max}$ of the node with the highest centrality and the centrality of another point V_i. The calculation method is shown in formula (2).

$$Center_D = \frac{1}{N-2}\sum_{i=1}^{N}[C_D(V_{\max}) - C_D(V_i)] \tag{2}$$

Among them, $V_{\max}$represents the centrality value of the node with the highest centrality in the network, and $C_D(V_i)$ represents the centrality of a certain node V_i, whose value is the ratio of the number of nodes k_i connected to it to the maximum possible number of nodes *N-1*. The calculation method is as follows.

$$C_D(V_i) = k_i/(N-1) \tag{3}$$

The more central the position of a node in the network, the stronger its control over information, and the lower the efficiency of information sharing in the network. This paper takes the opposite of the network centrality index, and the larger its value, the higher the efficiency of information sharing.

3.2.3. Enterprise risk-taking level

This paper uses the volatility of corporate earnings to measure the risk-taking level of listed companies. Higher risk-taking means increased uncertainty about a company's future cash inflows, and corporate earnings volatility is widely used to measure risk-taking. Roa_i represents the ratio of enterprise *i*'s profit before interest, taxes, depreciation and amortization (*EBITDA*) in the corresponding year to the total assets at the end of the year. We firstly adjust the company's *Roa* for each year using the industry average, and then

Table 4
Regression results of replacing measures of corporate risk-taking level.

	(1) roa range of three-year	(2) roa range of three-year	(3) standard deviation of five-year	(4) standard deviation of five-year	(5) roa rang of five-year	(6) roa rang of five-year
Density	0.0094***		0.0053***		0.0134***	
	(3.21)		(3.05)		(3.22)	
Center		0.0122***		0.0048**		0.0121**
		(3.57)		(2.37)		(2.50)
Size	−0.0012	−0.0011	−0.0013**	−0.0013*	−0.0038**	−0.0037**
	(−1.13)	(−1.05)	(−1.99)	(−1.90)	(−2.35)	(−2.27)
Lev	0.0054	0.0052	0.0003	0.0002	−0.0009	−0.0008
	(1.20)	(1.16)	(0.10)	(0.08)	(−0.13)	(−0.11)
Top1	−0.0003***	−0.0003***	−0.0002***	−0.0002***	−0.0006***	−0.0006***
	(−5.41)	(−5.33)	(−6.70)	(−6.52)	(−6.71)	(−6.51)
Salary	0.0011	0.0011	0.0003	0.0001	0.0006	0.0002
	(0.90)	(0.92)	(0.32)	(0.12)	(0.32)	(0.11)
TobinQ	0.0030***	0.0030***	0.0005	0.0004	0.0010	0.0008
	(6.62)	(6.68)	(1.61)	(1.36)	(1.47)	(1.21)
ROA	0.0225	0.0231	0.0096	0.0096	0.0366	0.0361
	(1.09)	(1.12)	(0.77)	(0.77)	(1.22)	(1.20)
Age	0.0041	0.0037	0.0001	−0.0003	0.0002	−0.0006
	(1.54)	(1.41)	(0.04)	(−0.16)	(0.05)	(−0.14)
Growth	0.0047***	0.0047***	0.0018**	0.0018**	0.0044**	0.0042**
	(3.61)	(3.63)	(2.37)	(2.29)	(2.38)	(2.29)
PPE	−0.0099*	−0.0095	−0.0227***	−0.0224***	−0.0545***	−0.0536***
	(−1.67)	(−1.61)	(−5.89)	(−5.86)	(−5.91)	(−5.85)
Cap	−0.0025***	−0.0025***	−0.0011***	−0.0011***	−0.0024**	−0.0025***
	(−3.82)	(−3.83)	(−2.79)	(−2.85)	(−2.51)	(−2.58)
Eps	−0.0030***	−0.0030***	−0.0025***	−0.0025***	−0.0056***	−0.0058***
	(−3.62)	(−3.57)	(−4.20)	(−4.33)	(−4.04)	(−4.17)
Dual	0.0014	0.0016	0.0010	0.0011	0.0021	0.0023
	(0.98)	(1.12)	(1.14)	(1.22)	(0.98)	(1.04)
Constant	0.0993***	0.1050***	0.0927***	0.0987***	0.2326***	0.2475***
	(4.33)	(4.66)	(5.95)	(6.41)	(6.23)	(6.71)
Ind	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
N	8428	8513	7425	7504	7425	7504
R^2	0.0645	0.0626	0.0814	0.0783	0.0829	0.0795

calculate the company's industry-adjusted *Roa* standard deviation during each observation period.

$$Risk_i = \sqrt{\frac{1}{N-1}\sum_{n=1}^{N}(Adj_Roa_{in} - \frac{1}{N}\sum_{n=1}^{N}Adj_Roa_{in})|N=3} \tag{4}$$

$$Adj_Roa_{in} = \frac{Ebitda_{in}}{Assets_{in}} - \frac{1}{X_n}\sum_{k=1}^{X}\frac{Ebitda_{kn}}{Assets_{kn}} \tag{5}$$

3.2.4. Information asymmetry

Referring to the method of Ascioglu et al. (2005), the KV measurement method is used to measure the degree of information asymmetry of listed companies. The KV measurement method measures the information disclosure quality through the slope of the regression of stock return on trading volume. The basic principle is that if the information disclosure quality of listed companies is poor, investors mainly obtain information based on the trading volume of listed companies' stocks, and have a low degree of dependence on information disclosure. At this time, changes in trading volume will cause large changes in stock prices, so the rate of change is an inverse indicator of information disclosure quality, that is, the smaller the KV value, the better the information disclosure quality. And the better the quality of information disclosure, the lower the degree of information asymmetry.

3.2.5. Analyst herding behavior

Referring to the method of Gleason and Lee (2003), the correlation between analysts' forecast changes in future *EPS* of the target company at different time points and the average forecast data of the entire industry analyst group is used as proxy indicator of analyst herding behavior. The formula is shown as follows.

$$Herd_{i,t} = (\sum_{j=1}^{N_i} Herd_analyst_{j,t})/N_i \tag{6}$$

Table 5
Regression results of replacing network density index.

	(1) Risk	(2) roa range of three-year	(3) standard deviation of five-year	(4) roa rang of five-year
New_Den	0.0012***	0.0025***	0.0013***	0.0033***
	(3.10)	(3.35)	(2.98)	(3.10)
Size	−0.0008	−0.0017	−0.0016**	−0.0044***
	(−1.52)	(−1.64)	(−2.35)	(−2.71)
Lev	0.0041*	0.0073	0.0004	−0.0004
	(1.67)	(1.60)	(0.15)	(−0.06)
Top1	−0.0001***	−0.0003***	−0.0002***	−0.0006***
	(−5.10)	(−5.20)	(−6.42)	(−6.44)
Salary	0.0005	0.0009	0.0001	0.0004
	(0.73)	(0.70)	(0.16)	(0.18)
TobinQ	0.0015***	0.0028***	0.0004	0.0008
	(6.29)	(6.38)	(1.30)	(1.13)
ROA	0.0061	0.0228	0.0088	0.0346
	(0.55)	(1.10)	(0.70)	(1.15)
Age	0.0018	0.0034	−0.0003	−0.0006
	(1.22)	(1.27)	(−0.15)	(−0.12)
Growth	0.0028***	0.0052***	0.0020**	0.0047**
	(3.90)	(3.92)	(2.52)	(2.49)
PPE	−0.0049	−0.0090	−0.0224***	−0.0540***
	(−1.50)	(−1.49)	(−5.74)	(−5.79)
Cap	−0.0015***	−0.0027***	−0.0012***	−0.0025***
	(−4.16)	(−4.06)	(−2.85)	(−2.61)
Eps	−0.0017***	−0.0029***	−0.0024***	−0.0055***
	(−3.69)	(−3.50)	(−4.04)	(−3.89)
Dual	0.0006	0.0012	0.0007	0.0013
	(0.83)	(0.80)	(0.75)	(0.58)
Constant	0.0667***	0.1265***	0.1049***	0.2617***
	(5.41)	(5.52)	(6.72)	(7.00)
Ind	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	8210	8210	7225	7225
R^2	0.0566	0.0579	0.0754	0.0770

where the core variable is $Herd_analyst_{j,t}$, which means an analyst will adjust the *EPS* forecast based on the overall industry judgment. That is, whether the analyst's last *EPS* forecast for stock *i* in year *t* has been adjusted in the direction of the average of the latest industry forecast compared to the first forecast of that year. If so, the value of *Herd_analys* is 1, otherwise it is 0. N_i is the total number of stock analyst teams that continue to track and predict company *i* for year *t*. When $Herd_index_{i,t}$ is larger, the degree of herdization predicted by securities analysts for company *i* is higher.

3.2.6. Control variables

Considering other factors that may affect the risk-taking level of listed companies, referring to the research of John et al. (2008) and Boubakri et al. (2013), the control variables include company size (*Size*), asset-liability ratio (*Lev*), salary sensitivity (*Salary*), shareholding ratio of the largest shareholder (*Top*1), company performance (*TobinQ*), return on assets (*ROA*), company operating years (*Age*), corporate sales revenue growth rate (*Growth*), fixed asset ratio (*PPE*), capital expenditures (*Cap*), earnings per share (*Eps*) and dual positions (*Dual*).

3.3. Model specification

In order to verify the relationship between institutional investors' information sharing and corporate risk taking, this paper constructed the following model.

$$Risk_{i,t} = \beta_0 + \beta_1 Ins_{i,t} + \beta Controls_{i,t} + \sum Ind_j + \sum Year_t + \varepsilon_{i,t} \tag{7}$$

where *Risk* represents the enterprise's risk-taking level; *Ins* represents network density (*Density*) and network centrality (*Center*); Controls represents a series of control variables selected above. In addition, in order to eliminate the impact of industry characteristics and time trends on the results, we also control industry (*Ind*) and year (*Year*) fixed effects (Table 1).

4. Empirical results

4.1. Data and descriptive statistical analysis

This paper selects the annual data of Shanghai and Shenzhen A-share listed companies from 2010 to 2022 as the research samples.

Table 6
Regression results of reducing the sample interval.

	(1) Risk	(2) Risk	(3) Risk	(4) Risk
Density	0.0034**	0.0044**		
	(1.98)	(2.29)		
Center			0.0037*	0.0050**
			(1.86)	(2.29)
Size		0.0007		0.0007
		(1.02)		(1.10)
Lev		0.0001		−0.0002
		(0.02)		(−0.07)
Top1		−0.0002***		−0.0002***
		(−5.90)		(−5.89)
Salary		0.0005		0.0005
		(0.58)		(0.60)
TobinQ		0.0012***		0.0012***
		(3.88)		(3.89)
ROA		0.0206		0.0210
		(1.59)		(1.62)
Age		0.0012		0.0011
		(0.67)		(0.64)
Growth		0.0023**		0.0023**
		(2.47)		(2.52)
PPE		−0.0053		−0.0051
		(−1.36)		(−1.33)
Cap		−0.0020***		−0.0019***
		(−4.53)		(−4.52)
Eps		−0.0015***		−0.0016***
		(−3.04)		(−3.09)
Dual		0.0009		0.0011
		(0.97)		(1.16)
Constant	0.0239***	0.0423***	0.0267***	0.0446***
	(2.81)	(2.72)	(3.16)	(2.93)
Ind	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	7065	6190	7142	6254
R^2	0.0503	0.0625	0.0484	0.0601

The data on the heavy holdings of institutional investors in this paper comes from the WIND database, and the company's financial data and other data come from the CSMAR database. We perform the following preprocessing on the original sample data: 1) Remove samples of financial listed companies; 2) Remove ST listed companies; 3) Remove samples with missing important variables. At the same time, in order to avoid the influence of outliers, all continuous variables are winsorized by 1 % above and below. Finally, 9515 firm-year sample observations were obtained.

Table 2 reports the descriptive statistical results of each main variable. It can be found that the maximum and minimum values of the enterprise's risk-taking level are 0.1699 and 0.0017 respectively, and there is a larger difference between them. The mean value of the enterprise's risk-taking level is 0.0272, which is greater than the median 0.0182, and the 75 % quantile and the 90 % quantiles are 0.0328 and 0.0579 respectively. The above data shows that there are large differences in the risk-taking levels of listed companies in China, and most companies tend to bear lower risks. For the measurement index of institutional investor network, the maximum and minimum values of network density are 1 and 0.1039 respectively. The network density of the institutional investor network with the highest network density is 41.67 % higher than that of other institutional investor networks, which illustrates that there is a difference in the speed of information transmission in the institutional investor network. And the maximum and minimum values of network centrality are 0 and −0.7180 respectively. The network centrality of the institutional investor network with the largest network centrality is 29.75 % higher than that of other institutional investor networks, indicating the difference in the position of institutional investors in the network. Overall, the efficiency of information sharing varies greatly among different institutional investor networks.

4.2. Analysis of regression results

Table 3 reports the benchmark regression results between institutional investor information sharing and corporate risk-taking. It can be found from columns (1) and (2) in the table that when no control variables are added, the influence coefficients of network density and network centrality of institutional investor networks on corporate risk-taking level are 0.0042 and 0.0048 respectively, and all are significant at the 1 % level. It indicates that without considering other control variables, institutional investor information sharing can significantly affect corporate risk-taking level. Taking control variables into account, the regression results in columns (3) and (4) illustrate that the impact of network density and network centrality on corporate risk-taking level is significantly positive at the 1 % level. That is, the higher the efficiency of institutional investor information sharing, the higher the risk-taking level of enterprises. As a result, the H1 hypothesis is illustrated.

Table 7
First-order difference regression results of institutional investor network indicators.

	(1) Risk	(2) Risk	(3) Risk	(4) Risk
d_Density	0.0029*	0.0037**		
	(1.79)	(2.17)		
d_Center			0.0053***	0.0066***
			(3.20)	(3.84)
Size		−0.0024***		−0.0024***
		(−3.07)		(−3.15)
Lev		0.0029		0.0024
		(0.85)		(0.69)
Top1		−0.0002***		−0.0002***
		(−4.35)		(−4.30)
Salary		0.0010		0.0009
		(1.10)		(1.07)
TobinQ		0.0006**		0.0006**
		(2.03)		(2.03)
ROA		0.0059		0.0044
		(0.39)		(0.29)
Age		0.0028		0.0028
		(1.32)		(1.34)
Growth		−0.0000		−0.0000
		(−0.04)		(−0.03)
PPE		−0.0120***		−0.0118***
		(−2.62)		(−2.59)
Cap		−0.0005		−0.0005
		(−0.99)		(−0.97)
Eps		−0.0019***		−0.0018***
		(−3.33)		(−3.23)
Dual		0.0017		0.0016
		(1.60)		(1.56)
Constant	0.0177**	0.0641***	0.0180**	0.0653***
	(2.12)	(3.69)	(2.15)	(3.78)
Ind	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	5834	5293	5896	5344.
R^2	0.0656	0.0776	0.0666	0.0785

Table 8
The regression results of instrumental variable.

	The first stage (1)	The second stage (2)
	Center	Risk
Center		0.1259***
		(2.35)
Turnover	0.0044***	
	(8.96)	
Constant	0.3301***	−0.7970
	(5.85)	(−0.26)
	Yes	Yes
Ind	Yes	Yes
Year	Yes	Yes
N	8210	8210
F	80.3336(P value=0.0098)	
Adj-R^2	0.1300	
Hansen J	12.7530	
Chi-sq (1) P-val	0.0004	

4.3. Robustness analysis

In order to test the robustness of the previous results, this paper investigates from the aspects of replacing the explained variable, changing the measurement method of the explanatory variable and adjusting the sample interval.

1) Replace the explained variable.

First, this paper uses the *Roa* range (*roa range of three-year*) of the company during the observation period to measure the volatility of earnings. Second, we take 5 years as an observation period and calculate the industry-adjusted *Roa* standard deviation (*standard deviation of five-year*) and range (*roa rang of five-year*) within each observation period. As shown in Table 4, the regression results have

Table 9
Mediating effect of information asymmetry.

	(1) Asy	(2) Asy
Density	−0.0921***	
	(−6.28)	
Center		−0.1171***
		(−6.84)
Size	0.0280***	0.0283***
	(6.96)	(7.12)
Lev	−0.0233	−0.0234
	(−1.25)	(−1.26)
Top1	−0.0001	−0.0002
	(−0.78)	(−1.09)
Salary	0.0140***	0.0137***
	(2.97)	(2.95)
TobinQ	0.0368***	0.0367***
	(17.70)	(17.82)
ROA	−0.1346	−0.1312
	(−1.50)	(−1.48)
Age	−0.0062	−0.0061
	(−0.67)	(−0.66)
Growth	0.0099	0.0075
	(1.37)	(1.05)
PPE	−0.0526**	−0.0501**
	(−2.21)	(−2.13)
Cap	0.0033	0.0036
	(1.12)	(1.23)
Eps	0.0499***	0.0498***
	(14.92)	(14.97)
Dual	−0.0080	−0.0091
	(−1.39)	(−1.58)
Constant	−0.5505***	−0.6385***
	(−6.47)	(−7.84)
Ind	Yes	Yes
Year	Yes	Yes
N	5589	5646
R^2	0.2963	0.2975

not changed substantially, so the conclusions of this paper are relatively robust.

2) Change the measure of network density. The network density of a stock is a decreasing function of the stock's market capitalization (Nagel, 2005). In order to further eliminate the impact of stock market capitalization on the results, the residuals after the regression of network density on stock circulation market capitalization are used as proxy variables for network density, and then the regression analysis is re-run. The results are shown in Table 5. After eliminating the impact of stock market capitalization, the relationships between network density and the standard deviation (*Risk*) and range (*roa range of three-year*) of *Roa* with a 3-year observation period and the standard deviation (*standard deviation of five-year*) and range (*roa rang of five-year*) of the *Roa* with a 5-year observation period are significantly positive, indicating that the conclusions of this paper is robust.

3) Reduce the sample interval. The occurrence of extreme events may affect the research conclusions of this paper. Considering the impact of the 2015 stock market crash and its lag, we eliminate the data from 2015 to 2016 and rerun the regression. The results are shown in Table 6. The relationships between network density and network centrality and corporate risk-taking levels are still significantly positive, indicating the robustness of the conclusions of this paper.

4) The method of first-order difference. In the regression model, it is necessary to consider such a mutually causal endogeneity problem, that is, companies with a high level of risk-taking are more likely to attract the attention of institutional investors, making the network density of the institutional investor networks larger. So, the increase in corporate risk-taking level is not due to the increase in the density of institutional investor networks, but to the large number of institutional investors holding shares in companies with higher risk-taking level. In order to control the problem of mutual causation, this paper uses institutional investor network indicators lagged one year to conduct regression analysis again. The test results are shown in Table 7. It can be seen from the regression results that the regression coefficients of network density and network centrality are both significantly positive, indicating that the positive correlation between institutional investor information sharing and corporate risk-taking level is not due to the endogenous relationship of reverse causation. Therefore, the above conclusions of this paper are robust.

5) The method of instrumental variable. Unobserved factors may cause institutional investors to invest in companies with higher level of risk-taking, biasing the estimates in this paper. Therefore, referring to the practice of Aggarwal et al. (2011), this paper selects stock turnover rate (*Turnover*) as an instrumental variable. Because institutional investors may hold shares of companies with high turnover rates, but turnover rates have no direct relationship with corporate risk-taking behavior, the condition of exogeneity of instrumental variables is met. The results are shown in Table 8. Column (1) is the first-stage regression result. The coefficient of *Turnover* is significantly positive, and the F value is greater than 10, indicating that the *Turnover* can affect the efficiency of information

Table 10
Mediating effect of analyst herding behavior.

	(1) Herd	(2) Herd
Density	−0.0699***	
	(−4.80)	
Center		−0.0976***
		(−5.83)
Size	−0.0048	−0.0044
	(−0.99)	(−0.92)
Lev	0.0012	0.0028
	(0.06)	(0.13)
Top1	0.0002	0.0001
	(0.69)	(0.57)
Salary	0.0110*	0.0116**
	(1.95)	(2.06)
TobinQ	0.0010	0.0008
	(0.43)	(0.36)
ROA	0.3858***	0.4064***
	(3.86)	(4.07)
Age	−0.0164	−0.0181
	(−1.32)	(−1.47)
Growth	0.0217***	0.0194***
	(3.03)	(2.72)
PPE	−0.0017	0.0030
	(−0.06)	(0.11)
Cap	0.0090***	0.0085***
	(2.80)	(2.65)
Eps	0.0115***	0.0108***
	(3.09)	(2.91)
Dual	−0.0044	−0.0051
	(−0.66)	(−0.77)
Constant	0.1663	0.0989
	(1.51)	(0.93)
Ind	Yes	Yes
Year	Yes	Yes
N	5630	5681
R^2	0.0932	0.0939

sharing among institutional investors. Column (2) shows that after controlling for the impact of possible omitted variables, the coefficient of network center is significantly positive, indicating that the effect of information sharing of institutional investors on improving corporate risk-taking level still exists.

5. Mechanism test

The above test results show that information sharing among institutional investors can improve corporate risk-taking level. This paper further attempts to explore the mechanism by which information sharing of institutional investor affects corporate risk-taking level. We use the following empirical model to test the mediating role of information asymmetry and analyst herding behavior:

$$Asy_{i,t} = \alpha_0 + \alpha_1 Ins_{i,t} + \alpha Controls_{i,t} + \sum Ind_j + \sum Year_t + \varepsilon_{i,t} \tag{8}$$

$$Herd_{i,t} = \gamma_0 + \gamma_1 Ins_{i,t} + \gamma Controls_{i,t} + \sum Ind_j + \sum Year_t + \varepsilon_{i,t} \tag{9}$$

where *Asy* represents the degree of information asymmetry, *Herd* represents analyst herding behavior, *Ins* represents network density and network centrality, and the remaining variables are consistent with the baseline model. If the signs of α_1 and γ_1 are significantly negative, it means that information asymmetry and analyst herding behavior play the mediating role in the impact of institutional investors' information sharing on corporate risk-taking level.

5.1. The role of information asymmetry

Table 9 shows the test results of the mediating effect of information asymmetry. Columns (1) and (2) show that the impact of network density and network centrality on information asymmetry is significantly negative at the 1 % levels, indicating that institutional investor information sharing can reduce the degree of corporate information asymmetry. Reducing the degree of information asymmetry helps external investors understand the company's situation more truly and reduces the financing constraints faced by the company, which can improve the company's risk-taking level. Therefore, information sharing by institutional investors can improve the level of corporate risk-taking by reducing the degree of corporate information asymmetry, that is, information asymmetry plays an

Table 11
Grouped regression results of financialization degree.

	High degree of financialization (1)	High degree of financialization (2)	Low degree of financialization (3)	Low degree of financialization (4)
Density	0.0075***		0.0027	
	(3.42)		(1.18)	
Center		0.0074***		0.0048*
		(2.92)		(1.79)
Lev	0.0038	0.0035	0.0002	0.0006
	(1.13)	(1.04)	(0.06)	(0.16)
Top1	−0.0002***	−0.0002***	−0.0001***	−0.0001***
	(−4.75)	(−4.59)	(−2.91)	(−2.90)
Salary	0.0005	0.0003	0.0008	0.0009
	(0.50)	(0.34)	(0.88)	(1.03)
TobinQ	0.0015***	0.0015***	0.0016***	0.0016***
	(4.58)	(4.56)	(4.33)	(4.34)
ROA	0.0075	0.0104	−0.0038	−0.0042
	(0.48)	(0.66)	(−0.24)	(−0.27)
Age	0.0031	0.0028	0.0019	0.0018
	(1.54)	(1.40)	(0.96)	(0.89)
Growth	−0.0006	−0.0005	0.0047***	0.0046***
	(−0.53)	(−0.42)	(4.97)	(4.92)
PPE	−0.0051	−0.0052	−0.0081*	−0.0071*
	(−1.06)	(−1.10)	(−1.86)	(−1.65)
Cap	−0.0008*	−0.0009*	−0.0017***	−0.0017***
	(−1.70)	(−1.80)	(−3.30)	(−3.41)
Eps	−0.0017***	−0.0017***	−0.0015**	−0.0015**
	(−2.75)	(−2.80)	(−2.34)	(−2.23)
Dual	0.0023**	0.0024**	−0.0001	0.0000
	(2.19)	(2.25)	(−0.11)	(0.01)
Constant	0.0593***	0.0643***	0.0526***	0.0535***
	(3.39)	(3.74)	(3.00)	(3.12)
Ind	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	4568	4601	3860	3912
R^2	0.0800	0.0749	0.0529	0.0515

intermediary role in the process of institutional investor information sharing affecting corporate risk-taking level. Hypothesis H2 has been verified.

5.2. The role of analyst herding behavior

Table 10 shows the test results of the mediating effect of analyst herding behavior. Columns (1) and (2) show that the effects of network density and network centrality on analyst herding behavior are both significantly negative at the 1 % level, indicating that institutional investor information sharing can inhibit analyst herding behavior. The reduction of analysts' herding behavior increases the company-related information obtained by external investors, which is beneficial to companies' acquisition of resources. At the same time, analysts' increased attention to the company can reduce management's opportunistic behavior, thereby improving the level of corporate risk-taking. Therefore, institutional investor information sharing can improve corporate risk-taking levels by inhibiting analyst herding behavior. That is, analyst herding behavior plays a mediating role in the process of institutional investor information sharing affecting corporate risk-taking levels. Hypothesis H3 has been verified.

6. Heterogeneity analysis

The previous analysis and empirical results show that information sharing in institutional investor networks can produce external supervision effect and resource advantage effect, which prompt managers to carry out high-risk investment activities that are more beneficial to enterprise development, and provide enterprises with corresponding resources support, thereby improving the level of corporate risk-taking. However, a company's operating activities and investment strategies will also be affected by the company's internal governance mechanism and the external environment. For companies with different characteristics, the impact of institutional investor networks on corporate risk-taking levels may be different. In consequence, based on the previous analysis, this paper further studies the heterogeneity of institutional investor information sharing affecting corporate risk-taking levels from the perspectives of corporate internal governance and external environment.

6.1. The effect of financialization

In recent years, China's economy has experienced a trend of financialization. Generally speaking, corporate investment in financial

Table 12
Grouped regression results of differences in financing constraints.

	High degree of financing constraints (1)	High degree of financing constraints (2)	Low degree of financing constraints (3)	Low degree of financing constraints (4)
Density	0.0045**		0.0033	
	(1.96)		(1.58)	
Center		0.0089***		0.0024
		(3.25)		(1.01)
Size	0.0012	0.0013	−0.0006	−0.0006
	(1.29)	(1.47)	(−0.80)	(−0.81)
Lev	0.0061*	0.0061*	−0.0063*	−0.0063*
	(1.91)	(1.92)	(−1.75)	(−1.75)
Top1	−0.0002***	−0.0002***	−0.0001**	−0.0001**
	(−4.95)	(−4.84)	(−2.38)	(−2.37)
Salary	−0.0001	−0.0000	0.0018**	0.0017**
	(−0.16)	(−0.03)	(2.19)	(2.06)
TobinQ	0.0018***	0.0019***	0.0009**	0.0008**
	(5.56)	(5.86)	(2.30)	(2.19)
ROA	0.0139	0.0142	0.0508***	0.0511***
	(0.90)	(0.93)	(3.08)	(3.11)
Age	0.0029	0.0025	0.0006	0.0006
	(1.58)	(1.35)	(0.29)	(0.32)
Growth	−0.0012	−0.0012	0.0059***	0.0060***
	(−0.84)	(−0.91)	(7.28)	(7.38)
PPE	−0.0103**	−0.0103**	0.0019	0.0024
	(−2.25)	(−2.25)	(0.46)	(0.58)
Cap	−0.0015***	−0.0014***	−0.0015***	−0.0015***
	(−2.88)	(−2.83)	(−3.22)	(−3.25)
Eps	−0.0066***	−0.0065***	0.0013***	0.0013***
	(−8.64)	(−8.50)	(2.63)	(2.59)
Dual	0.0009	0.0011	0.0021*	0.0020*
	(0.91)	(1.10)	(1.94)	(1.89)
Constant	0.0282	0.0274	0.0362**	0.0401**
	(1.44)	(1.43)	(1.97)	(2.21)
Ind	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	4616	4674	3812	3839
R^2	0.1069	0.1064	0.0501	0.0500

products will have two opposite effects on corporate development. One is that enterprises use part of their idle funds to invest in short-term financial products, which can increase the liquidity of corporate assets and, to a certain extent, alleviate the constraints on the company's main business investment activities caused by future shortages of funds. The other is that the high yield of financial products causes companies to allocate too much financial assets, crowding out funds for investment in other main businesses, resulting in underinvestment of corporates. However, both of these effects will have an impact on the company's future investment activities, thereby affecting the company's risk-taking level. Therefore, this paper selects the ratio of financial assets held by enterprises to total assets to measure the degree of financialization of enterprises according to the method of Orhangazi (2008) and Demir (2009), and divides the samples into low financialization and high financialization according to the median level. We examine the difference in the impact of institutional investor information sharing on corporate risk-taking level in samples of enterprises with different degrees of financialization.

The regression results are shown in Table 11. It can be seen that in the sample of enterprises with a high degree of financialization, the coefficient of network density is significantly positive at the 1 % level, while in the sample of enterprises with a low degree of financialization, the coefficient of network density is not significant. In addition, in the sample of enterprises with a high degree of financialization, the coefficient of network centrality is significantly positive at the 1 % level, but in the sample of enterprises with a low degree of financialization, the coefficient of network centrality is significantly positive at the 10 % level. Overall, in enterprises with a high degree of financialization, the impact of information sharing by institutional investors on the level of corporate risk-taking is more significant. The above results may be due to the fact that enterprises with a higher degree of financialization face higher resource constraints when conducting other investment activities, so the resource advantage effect of institutional investment networks can play a greater role. At the same time, the external supervision effect of the institutional investor network can better restrain managers from over-allocating financial assets. It prompts managers to carry out investment activities that are more conducive to the long-term development of enterprises, thereby improving the level of corporate risk-taking.

6.2. Perspective of enterprise resource acquisition

6.2.1. Financing constraints

Financing constraints mean a company can obtain sufficient funds for investment in high-risk and high-yield projects, which will

Table 13
Regression results of political connections.

	Have political connections (1)	Have political connections (2)	No political connections (3)	No political connections (4)
Density	0.0021		0.0059***	
	(0.77)		(2.98)	
Center		0.0015		0.0082***
		(0.47)		(3.67)
Size	−0.0004	−0.0006	−0.0009	−0.0007
	(−0.47)	(−0.64)	(−1.31)	(−1.08)
Lev	0.0033	0.0038	0.0036	0.0032
	(0.79)	(0.92)	(1.19)	(1.06)
Top1	−0.0002***	−0.0002***	−0.0001***	−0.0001***
	(−3.41)	(−3.41)	(−3.99)	(−3.87)
Salary	−0.0004	−0.0004	0.0003	0.0004
	(−0.35)	(−0.36)	(0.41)	(0.45)
TobinQ	0.0018***	0.0017***	0.0014***	0.0014***
	(3.96)	(3.90)	(4.79)	(4.89)
ROA	0.0121	0.0104	0.0078	0.0087
	(0.62)	(0.53)	(0.57)	(0.64)
Age	0.0013	0.0013	0.0035*	0.0033*
	(0.55)	(0.55)	(1.93)	(1.85)
Growth	0.0014	0.0014	0.0032***	0.0032***
	(1.03)	(1.06)	(3.80)	(3.86)
PPE	−0.0103*	−0.0089*	−0.0022	−0.0024
	(−1.92)	(−1.66)	(−0.55)	(−0.62)
Cap	−0.0013**	−0.0014**	−0.0012***	−0.0012***
	(−2.11)	(−2.20)	(−2.75)	(−2.74)
Eps	−0.0029***	−0.0028***	−0.0013**	−0.0013**
	(−3.45)	(−3.34)	(−2.46)	(−2.45)
Dual	−0.0026*	−0.0027**	0.0021**	0.0023**
	(−1.93)	(−2.01)	(2.23)	(2.48)
Constant	0.0727***	0.0783***	0.0486***	0.0494***
	(3.67)	(4.01)	(3.08)	(3.20)
Ind	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	2648	2673	5780	5840
R^2	0.0881	0.0865	0.0588	0.0571

affect the company's risk-taking level. From the previous theoretical analysis, it can be seen that corporate risk-taking is resource-consuming activity. If there are not enough resources, companies will face greater investment restrictions, which reduces the level of corporate risk-taking. The institutional investor network has a resource support function and can reduce the financing constraints faced by enterprises. Hence, this paper expects that the higher the corporate financing constraints, the stronger the effect of institutional investor networks on corporate risk-taking levels. We use the WW index as a proxy variable for corporate financing constraints according to the method of Whited and Wu (2006), divide the sample into high-financing-constrained and low-financing-constrained enterprises according to the median level, and then perform group regression.

The regression results are shown in Table 12. In companies with higher financing constraints, the coefficients of network density and network centrality are significantly positive at the 5 % and 1 % levels respectively. While in enterprises with lower financing constraints, the coefficients of network density and network centrality are not significant. These show that the resource support function of institutional investor networks can play a more valuable role for companies with higher financing constraints, and significantly improve the level of corporate risk-taking.

6.2.2. Political connections

By actively establishing ties with the government, companies can obtain government subsidies, tax incentives, and more bank loans for high-risk investment activities. This study shows that the information advantages of institutional investor networks can reduce corporate information asymmetry, help companies obtain more high-quality investment projects, and improve corporate resource acquisition capabilities. As a result, whether there is a connection with the government will affect the relationship between institutional investor networks and corporate risk taking. Based on the above analysis, referring to the research of Otchere et al. (2020), this paper takes whether the chairman or general manager hold a government position as the basis for whether a company is related to the government. And the heterogeneity of how institutional investor information sharing affects corporate risk-taking levels in different samples is examined.

Columns (1) and (2) in Table 13 show that in politically connected enterprises, the coefficients of network density and network centrality are not significant. Columns (3) and (4) show that in companies without political connections, the coefficients of network density and network centrality are both significantly positive at the 1 % level. The above results show that enterprises with connections to the government can obtain more government support, and have more funds available for high-risk investment activities. Among enterprises that are not related to the government, the enterprises face greater resource constraints, so the resource advantage effect of

Table 14
Grouped regression results of agency cost.

	High agency costs (1)	High agency costs (2)	Low agency costs (3)	Low agency costs (4)
Density	0.0054**		0.0025	
	(2.27)		(1.20)	
Center		0.0108***		0.0024
		(3.90)		(0.98)
Size	−0.0005	−0.0005	0.0005	0.0005
	(−0.66)	(−0.59)	(0.62)	(0.74)
Lev	0.0026	0.0027	0.0032	0.0028
	(0.78)	(0.80)	(0.89)	(0.81)
Top1	−0.0001***	−0.0001***	−0.0001***	−0.0001***
	(−3.69)	(−3.61)	(−2.99)	(−3.08)
Salary	0.0015	0.0016*	−0.0001	−0.0001
	(1.58)	(1.71)	(−0.06)	(−0.16)
TobinQ	0.0017***	0.0018***	0.0016***	0.0016***
	(5.29)	(5.50)	(4.37)	(4.40)
ROA	−0.0426**	−0.0420**	0.0637***	0.0643***
	(−2.45)	(−2.42)	(4.34)	(4.40)
Age	0.0036*	0.0029	0.0021	0.0022
	(1.88)	(1.53)	(1.03)	(1.09)
Growth	0.0023**	0.0024**	0.0030***	0.0030***
	(2.12)	(2.21)	(3.27)	(3.29)
PPE	−0.0115**	−0.0112**	−0.0012	−0.0010
	(−2.47)	(−2.43)	(−0.29)	(−0.23)
Cap	−0.0007	−0.0007	−0.0021***	−0.0021***
	(−1.33)	(−1.30)	(−4.56)	(−4.56)
Eps	−0.0047***	−0.0046***	−0.0004	−0.0004
	(−5.86)	(−5.76)	(−0.72)	(−0.73)
Dual	0.0012	0.0014	−0.0002	−0.0002
	(1.18)	(1.30)	(−0.19)	(−0.16)
Constant	0.0271	0.0301*	0.0551***	0.0562***
	(1.59)	(1.80)	(3.15)	(3.29)
Ind	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	4169	4213	4259	4300
R^2	0.0849	0.0838	0.0562	0.0560

the institutional investor network plays a greater marginal role and can significantly improve the risk-taking level of the enterprise.

6.3. *Perspective of internal corporate governance*

6.3.1. *Agency costs*

Under the system of separation of ownership and management rights, principal-agent problems are common in enterprises. Corporate owners with ownership rights and managers with management rights have inconsistent utility functions, resulting in managers not necessarily making investment decisions based on shareholders' demands and wealth maximization. Companies with more serious agency problems have managers with more decision-making autonomy and more room for opportunistic behavior, which may affect the level of corporate risk-taking. Based on the above analysis, this paper uses the management expense ratio as the variable of corporate agency cost according to the method of Ang et al. (2000), divides the sample into enterprises with low agency cost and high agency cost according to the median level, and examines the differences in the impact of institutional investor information sharing on corporate risk-taking levels in corporate samples with different agency costs.

The regression results are shown in Table 14. The results in columns (1) and (2) of the table show that in enterprises with higher agency costs, the coefficients of network density and network centrality are significantly positive on the level at 5 % and 1 % respectively. The results in columns (3) and (4) of the table show that in enterprises with low agency costs, the coefficients of network density and network centrality are not significant. The above results show that the more severe the agency costs, the higher the possibility of corporate management to engage in opportunistic behavior, so the supervisory effect of institutional investors' information sharing can play a more significant role. That is, the higher the agency costs within a company, the more the institutional investor network can improve the level of corporate risk-taking.

6.3.2. *Internal control quality*

In the context of the separation of ownership and control in the corporate system, managers do not act entirely in accordance with the owners' business objectives. In their daily business activities, managers may avoid high-risk, high-return investment activities for their own interests, reputation, etc. In different corporate governance mechanisms, the decision-making behavior of management is different. At the same time, institutional investors, as external stakeholders, can supervise the behavior of management. The effectiveness of the external supervision mechanism also depends on the level of internal governance of the enterprise. Based on this, we

Table 15
Grouped regression results of internal control quality.

	Good quality of internal control (1)	Good quality of internal control (2)	Bad quality of internal control (3)	Bad quality of internal control (4)
Density	0.0058***		0.0018	
	(2.91)		(0.71)	
Center		0.0079***		0.0033
		(3.52)		(1.06)
Size	−0.0010	−0.0010	0.0001	0.0002
	(−1.60)	(−1.56)	(0.14)	(0.21)
Lev	−0.0036	−0.0039	0.0069*	0.0070*
	(−1.22)	(−1.32)	(1.85)	(1.89)
Top1	−0.0001**	−0.0001**	−0.0002***	−0.0002***
	(−2.09)	(−2.02)	(−4.75)	(−4.76)
Salary	0.0012	0.0012	0.0008	0.0008
	(1.51)	(1.53)	(0.75)	(0.80)
TobinQ	0.0014***	0.0014***	0.0018***	0.0018***
	(4.82)	(4.90)	(4.30)	(4.37)
ROA	0.0354***	0.0362***	−0.0154	−0.0162
	(2.65)	(2.72)	(−0.84)	(−0.88)
Age	0.0025	0.0024	0.0024	0.0022
	(1.51)	(1.45)	(1.14)	(1.04)
Growth	0.0064***	0.0064***	−0.0003	−0.0003
	(7.63)	(7.66)	(−0.26)	(−0.28)
PPE	−0.0061	−0.0067*	−0.0083*	−0.0068
	(−1.56)	(−1.75)	(−1.70)	(−1.40)
Cap	−0.0009**	−0.0009**	−0.0022***	−0.0022***
	(−2.06)	(−2.05)	(−3.95)	(−3.96)
Eps	0.0012**	0.0013**	−0.0059***	−0.0058***
	(2.45)	(2.51)	(−6.92)	(−6.79)
Dual	0.0014	0.0015*	−0.0001	−0.0000
	(1.51)	(1.66)	(−0.08)	(−0.01)
Constant	0.0439***	0.0484***	0.0567***	0.0563***
	(3.06)	(3.46)	(2.92)	(2.95)
Ind	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	4924	4962	3504	3551
R^2	0.0585	0.0587	0.0832	0.0815

divide the internal control quality of enterprises into high internal control quality and low internal control quality samples according to the median level, and examine the differences in the impact of institutional investors' information sharing on the level of corporate risk-taking in different samples. Drawing on the research of Zheng and Han, we use the Dibo Internal Control Index as an indicator to measure the quality of corporate control.

The regression results are shown in Table 15. The results in columns (1) and (2) of the table show that in companies with higher information disclosure quality, the coefficients of network density and network centrality are significantly positive on the level at 5 % and 1 % respectively. While the results in columns (3) and (4) of the table show that in companies with low information disclosure quality, the coefficients of network density and network centrality are not significant. The reason for this phenomenon may be that companies with poor information disclosure quality have a higher degree of information asymmetry and analysts' herding behavior is more serious, which inhibits the intermediary mechanism of institutional investors' information sharing affecting risk-taking level of corporate. Hence, companies with better information disclosure quality can ensure that institutional investor information sharing can affect corporate risk-taking levels by reducing corporate information asymmetry and inhibiting analyst herding behavior.

6.4. External environment perspective

6.4.1. Legal environment

The conclusion of this paper shows that information sharing among institutional investors will improve the efficiency of information dissemination, and reduce information asymmetry between internal and external enterprises, enhancing the risk-taking level of enterprises. However, whether regional regulations and systems are complete and whether judicial and administrative agencies are strictly enforcing laws will directly affect corporate governance behaviors. When the legal environment in the region where a company is located is poor, management is more likely to conceal poor performance and bad news, resulting in lower corporate information transparency and exacerbating information asymmetry between the company and external investors. Therefore, the impact of institutional investor information sharing on corporate risk-taking levels may differ under different legal environments. Based on the above analysis, we use the marketization index compiled by the Institute of National Economy as the proxy variable for the regional legal level according to the research of Fu et al. (2025), divides the samples into regions with good and poor rule of law levels according to the median level, and groups them to test the impact of information sharing by institutional investors on the level of corporate risk-taking.

Table 16
Grouped regression results of legal environment.

	Good legal environment (1)	Good legal environment (2)	Bad legal environment (3)	Bad legal environment (4)
Density	0.0073***		0.0017	
	(3.01)		(0.80)	
Center		0.0073**		0.0051**
		(2.55)		(2.12)
Size	−0.0000	−0.0000	−0.0009	−0.0008
	(−0.05)	(−0.02)	(−1.13)	(−1.08)
Lev	−0.0027	−0.0030	0.0061*	0.0062*
	(−0.76)	(−0.85)	(1.85)	(1.90)
Top1	−0.0001***	−0.0001***	−0.0001***	−0.0001***
	(−3.70)	(−3.65)	(−3.58)	(−3.59)
Salary	0.0014	0.0013	0.0001	0.0003
	(1.42)	(1.34)	(0.16)	(0.35)
TobinQ	0.0009***	0.0009**	0.0020***	0.0020***
	(2.63)	(2.55)	(5.66)	(5.86)
ROA	0.0050	0.0054	0.0078	0.0089
	(0.31)	(0.34)	(0.50)	(0.57)
Age	0.0011	0.0011	0.0021	0.0019
	(0.59)	(0.54)	(1.04)	(0.93)
Growth	0.0065***	0.0064***	−0.0002	−0.0000
	(5.84)	(5.79)	(−0.18)	(−0.05)
PPE	−0.0004	−0.0012	−0.0087**	−0.0077*
	(−0.08)	(−0.24)	(−2.04)	(−1.83)
Cap	−0.0016***	−0.0016***	−0.0013***	−0.0013***
	(−3.03)	(−3.00)	(−2.69)	(−2.68)
Eps	−0.0012**	−0.0012**	−0.0027***	−0.0027***
	(−2.02)	(−1.96)	(−4.09)	(−4.06)
Dual	0.0016	0.0016	0.0003	0.0005
	(1.51)	(1.52)	(0.26)	(0.47)
Constant	0.0360*	0.0418**	0.0660***	0.0650***
	(1.89)	(2.23)	(3.98)	(3.99)
Ind	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	3919	3945	4509	4568
R^2	0.0752	0.0757	0.0649	0.0629

The regression results are shown in Table 16. For enterprises in areas with good legal environment, the coefficient of network density is significantly positive at the 1 % level, while for enterprises in areas with poor legal environment, the coefficient of network density is not significant. In addition, for enterprises in areas with good legal environment, network centrality increases every 1 unit, the level of enterprise risk-taking increases by 0.0073 %. For enterprises in areas with poor legal environment, network centrality increases every 1 unit, the level of enterprise risk-taking increases by 0.0051 %. Overall, in areas with a good legal environment, institutional investor information sharing has a more significant impact on corporate risk-taking levels. This may be due to that a high level of legal rule makes corporate information, financial and other disclosure behaviors more standardized and complete, and corporate information transparency is higher. Hence, institutional investors play a greater role in improving corporate risk-taking levels by reducing information asymmetry.

6.4.2. Enterprise life cycle

According to the research of this paper and the theoretical analysis of previous literature, the financing constraint of enterprises is one of the main factors that restrict enterprises from taking risks. The theory of enterprise life cycle shows that enterprises in different life cycles have different characteristics in governance, operation, finance, etc. Therefore, there are differences in the financing needs and financing constraints faced by enterprises in different life cycles. Based on this, this paper believes that the impact of information sharing by institutional investors on the level of corporate risk-taking may vary depending on the different life cycles of enterprises. To test the above logic, this paper, based on Dickinson's approach (2011), uses a combination of operating cash flow, investment cash flow and financing cash flow to divide the life cycle of an enterprise into growth, maturity and decline stages, and further refers to the maturity and decline stages as the non-growth stage.

The regression results are shown in Table 17. Columns (1) and (2) show that when an enterprise is in the growth stage, the coefficients of network density and network centrality are significantly positive at the 1 % level. Columns (3) and (4) show that for enterprises in the non-growth stage, the coefficients of network density and network centrality are not significant. The above results show that the role of information sharing by institutional investors in improving the risk-taking level of enterprises is greater in the growth stage of enterprises, but smaller in the non-growth stage. This may be because non-growth-stage enterprises have stable profits and continuous cash flow, and stable internal financing can generally support their investment needs, so the financing constraints they face will not have a significant impact on the investment activities of enterprises. Growth-stage enterprises are developing rapidly and have greater financing needs, but at this time, the investment risks of enterprises are also higher, and the financing constraints they

Table 17
Grouped regression results of enterprise life cycle.

	Growth stage (1)	Growth stage (2)	Non-growth stage (3)	Non-growth stage (4)
Density	0.0058***		0.0018	
	(2.91)		(0.71)	
Center		0.0079***		0.0033
		(3.52)		(1.06)
Size	−0.0010	−0.0010	0.0001	0.0002
	(−1.60)	(−1.56)	(0.14)	(0.21)
Lev	−0.0036	−0.0039	0.0069*	0.0070*
	(−1.22)	(−1.32)	(1.85)	(1.89)
Top1	−0.0001**	−0.0001**	−0.0002***	−0.0002***
	(−2.09)	(−2.02)	(−4.75)	(−4.76)
Salary	0.0012	0.0012	0.0008	0.0008
	(1.51)	(1.53)	(0.75)	(0.80)
TobinQ	0.0014***	0.0014***	0.0018***	0.0018***
	(4.82)	(4.90)	(4.30)	(4.37)
ROA	0.0354***	0.0362***	−0.0154	−0.0162
	(2.65)	(2.72)	(−0.84)	(−0.88)
Age	0.0025	0.0024	0.0024	0.0022
	(1.51)	(1.45)	(1.14)	(1.04)
Growth	0.0064***	0.0064***	−0.0003	−0.0003
	(7.63)	(7.66)	(−0.26)	(−0.28)
PPE	−0.0061	−0.0067*	−0.0083*	−0.0068
	(−1.56)	(−1.75)	(−1.70)	(−1.40)
Cap	−0.0009**	−0.0009**	−0.0022***	−0.0022***
	(−2.06)	(−2.05)	(−3.95)	(−3.96)
Eps	0.0012**	0.0013**	−0.0059***	−0.0058***
	(2.45)	(2.51)	(−6.92)	(−6.79)
Dual	0.0014	0.0015*	−0.0001	−0.0000
	(1.51)	(1.66)	(−0.08)	(−0.01)
Constant	0.0439***	0.0484***	0.0567***	0.0563***
	(3.06)	(3.46)	(2.92)	(2.95)
Ind	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	4924	4962	3504	3551
R^2	0.0585	0.0587	0.0832	0.0815

face are greater. Therefore, information sharing by institutional investors can play a greater marginal effect to ease the financing constraints of enterprises, thereby improving the risk-taking level of enterprises.

7. Conclusion

Based on the social network theory, this paper constructs an institutional investor network by connecting institutional investors with heavy holdings of shares of the same listed company, and studies the impact of information sharing among institutional investors within the network on corporate risk-taking levels. The study found that information sharing among institutional investors can improve corporate risk-taking levels. The higher the efficiency of information sharing, the higher the risk-taking level of listed companies. In particular, this paper finds that institutional investor information sharing affects corporate risk-taking levels by reducing the degree of information asymmetry and reducing analyst herding behavior. In order to verify the robustness of the conclusion, we successively changed the calculation method of corporate risk-taking level, the measurement method of information sharing efficiency, the sample selection time range, and used the first-order lag test and instrumental variable method to control endogeneity. In addition, based on the internal and external perspectives of the enterprise, the difference in the network governance role of institutional investors under different circumstances was examined. The analysis found that in samples with high degree of financialization, high financing constraints, no political connections, high agency costs, good information disclosure quality, good legal environment, and relatively developed regional economies, The analysis found that in samples with a high degree of financialization, high financing constraints, no political connections, high agency costs, good internal control quality, good legal environment, and growth stage company, institutional investor information sharing can more significantly improve the enterprises risk-taking level.

Based on the above research conclusions and combined with the governance situation of listed companies in China, this paper puts forward the following suggestions: (1) Listed companies should actively introduce institutional investors, increase the shareholding ratio of institutional investors, so as to form a multi-dimensional institutional investor network and improve corporate risk-taking level. (2) Institutional investors should make full use of resources in the network to enhance their voice in participating in corporate governance. So institutional investors need to strengthen their connections with other institutional investors or market participants, and make up for the disadvantages caused by information asymmetry, as well as improving the ability to participate in the governance of listed companies. (3) Policymakers should further introduce relevant policies and measures to continuously improve the market environment, and provide a better market environment and institutional guarantee for the long-term development of

institutional investors. At the same time, accelerating market-oriented reform will help improve the risk-taking level of enterprises, thereby promoting the economic development of the entire society.

Author contributions

All authors have read and approved the final manuscript.

CRediT authorship contribution statement

Xiong Xiong: Formal analysis, Conceptualization, Funding acquisition, Data curation. **Ya-Li Dong:** Writing – original draft. **Xiao-Li Gong:** Writing – review & editing.

Declaration of Competing interest

The authors declare that we have no competing interests.

Acknowledgments

We would like to acknowledge the financial support from the National Natural Science Foundation of China (Nos. 72271135, 72141304, 71901130); National Social Science Fund of China (22&ZD117); Special Funds for Taishan Scholars (tsqn202211120); Laboratory of Computation and Analytics of Complex Management Systems (Tianjin University); 2024 Qingdao Social Science Planning Project. Qingdao University Interdisciplinary Joint Research Project (XT2024302).

Data availability

Data will be made available on request.

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